

BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

Metallic Bonding — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. Attraction between positive metal ions and delocalised electrons.
2. Delocalised electrons.
3. Can be hammered into sheets without breaking.

B

4. Lattice of positive ions surrounded by a sea of delocalised electrons.
5. Electrons transfer kinetic energy rapidly through the lattice.
6. Layers of ions slide; electrons still hold the structure together.

C

7. Mg has mobile electrons; MgO ions are fixed in the solid lattice.
8. Different-sized atoms in the alloy hinder layers sliding.
9. Only one delocalised electron per atom / 1+ ions → weaker metallic bonding.

D

10. Mg²⁺ lattice; delocalised electrons; electrostatic attraction is metallic bonding.
11. Metal layers slide and electrons still attract ions. Ionic lattice: sliding lines up like charges, so it shatters.
12. Delocalised electrons make it a good conductor; it is also ductile.

E

13. Al forms 3+ ions and contributes more delocalised electrons than Na (1+), so the metallic attraction is stronger and the melting point is higher.